Embedded Systems and Intelligent Infrastructure: A Field Guide

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About this resource

This original NI4E editorial resource supports professional learning and discussion. It is not represented as a peer-reviewed journal paper, formal engineering standard, or substitute for project-specific professional advice.

From sensor to decision

An intelligent system typically combines sensing, signal conditioning, processing, communications and an action or display. Good design starts by defining what must be measured, the acceptable error range, the environment, the sampling frequency and what decision the data should support.

Design considerations

Engineers should account for power supply stability, electrical noise, calibration, environmental protection, maintainability and failure behaviour. Communication loss and sensor drift must be considered in the design, not only after deployment.

Security and documentation

Connected devices should use controlled access, secure configuration, appropriate updates and minimal collection of personal or operational data. Documentation should record wiring, firmware versions, calibration, test results and maintenance procedures.

Applications

Examples include equipment-condition monitoring, energy metering, water-system monitoring, building controls and industrial process instrumentation. The most useful solution is one that operators can understand, maintain and trust.

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